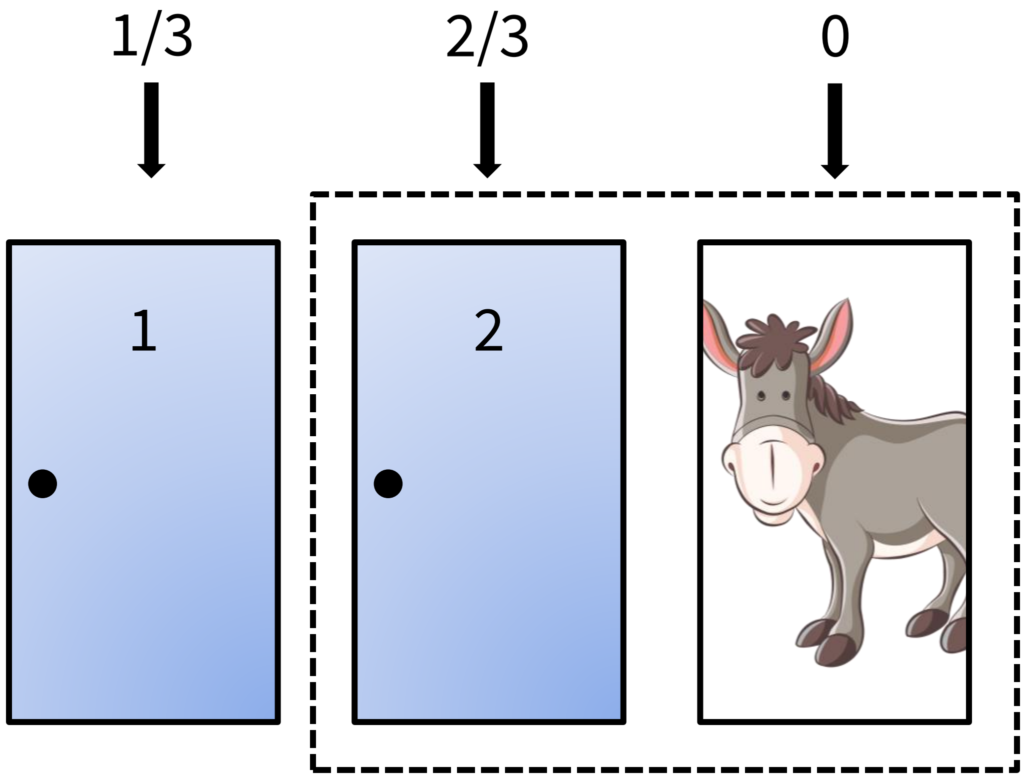
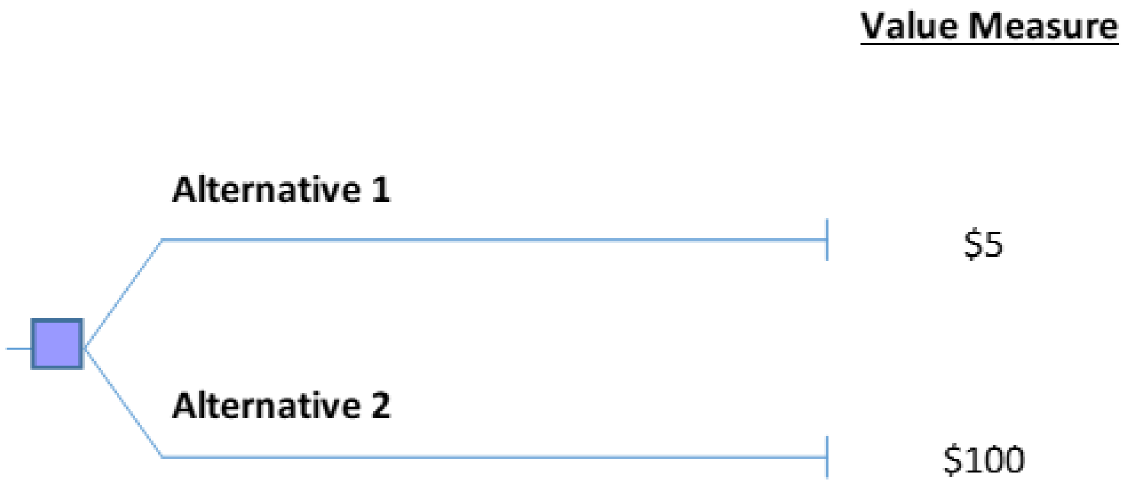
Monty Hall Problem Solution In Cartoon Form



Simple Decision

- How would you choose?



- This decision is uninteresting because it is a deterministic decision
- To deal with interesting decisions, we must be able to deal with ***uncertainty***

Probability

- Two kinds of uncertainty
  - Aleatory: fundamental, natural randomness in a process
    - Encoded by a probability density function (PDF)
  - Epistemic: scientific uncertainty in the model of the process
    - Encoded by alternate models and their PDFs
  - Example of the thumbtack
    - Aleatory uncertainty: how the tack lands heads/tails
    - Epistemic uncertainty: how we model the thumbtack and its behavior
- Two approaches to probability
  - Frequentists/classical statistics
  - Bayesian probability
- Sources of data in the Bayesian context
  - Statistics
  - Industry Reports
  - Test results
  - Expert opinions

Misperceptions on Probability

- (1) Real-world uncertainties are anything alike probability scenarios from a probability class.
- (2) Perfectly shuffled decks of cards and perfectly fair dice exist.
- (3) Probabilities are objective ideas, regardless of the perspective of the individual.

$$
\begin{aligned}
\mathbb{E}(X^2) &= \sum_{k=0}^{n} k^2 \binom{n}{k} p^k (1-p)^{n-k} = \sum_{k=1}^{n} (k(k-1)+k) \binom{n}{k} p^k (1-p)^{n-k} \\
&= \sum_{k=2}^{n} k(k-1) \binom{n}{k} p^k (1-p)^{n-k} + \underbrace{\sum_{k=1}^{n} k \binom{n}{k} p^k (1-p)^{n-k}}_{np} \\
&= n(n-1)p^2 \sum_{k=2}^{n} \binom{n-2}{k-2} p^{k-2} (1-p)^{n-2-(k-2)} + np \\
&= n(n-1)p^2 \sum_{j=0}^{n-2} \binom{n-2}{j} p^j (1-p)^{n-2-j} + np \\
&= n(n-1)p^2 + np = np((n-1)p+1) = np(np-p+1).
\end{aligned}
$$

Some Philosophical Thoughts on Probability

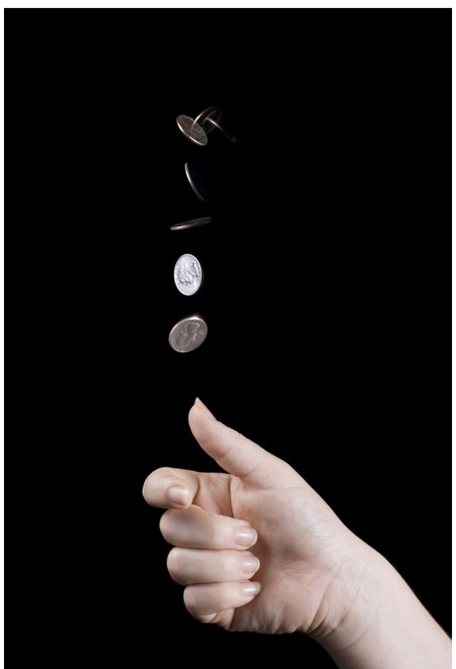
- **Question**: What is a probability?
- All probabilities are conditioned on the total sum of your life experiences and observed information: "&"
  - e.g. My belief on the probability of rain tomorrow $=P(rain \mid \&)$
  - & = my knowledge of seasons, coincidental factors, etc.
  - Your probability **may be different** from my probability on the same uncertainty
- Probabilities are a **belief**
- We should strive to become "Bayesian Thinkers"
  - We have the ability to observe a body of evidence, and update our beliefs based upon new evidence

Thomas Bayes
1701-1761

Key Takeaways on Probability

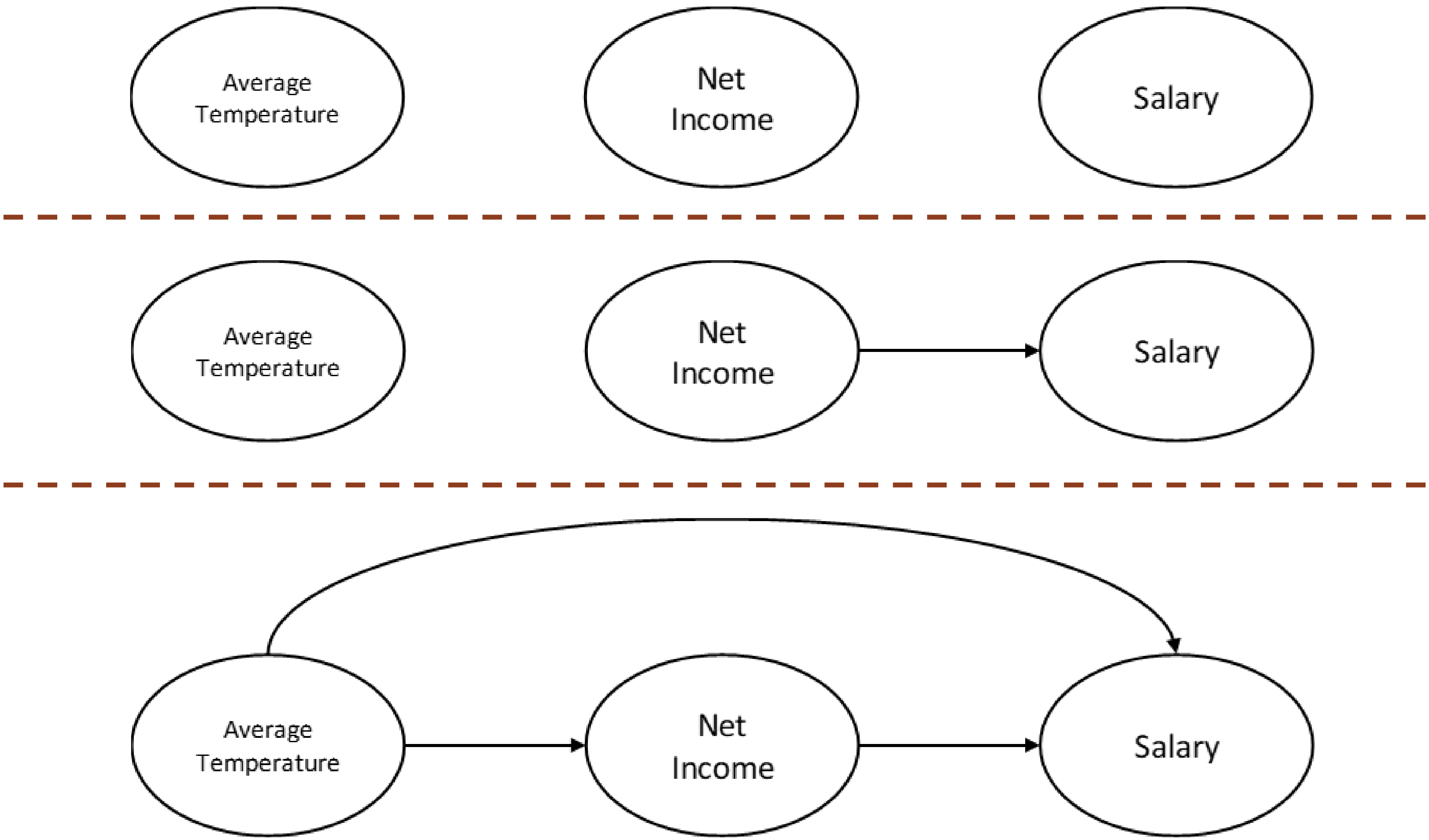
- Key point 1: **Everything in the universe is uncertain**
- Key point 2: **We should decouple decisions from their outcomes**

- Stochastic process: any event whose outcome is unknown
- The universe is a stochastic process



Probabilistic Dependence and Independence

- Question: What is your decision frame?

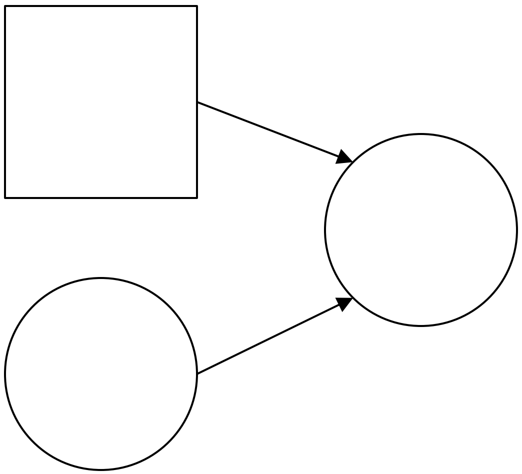


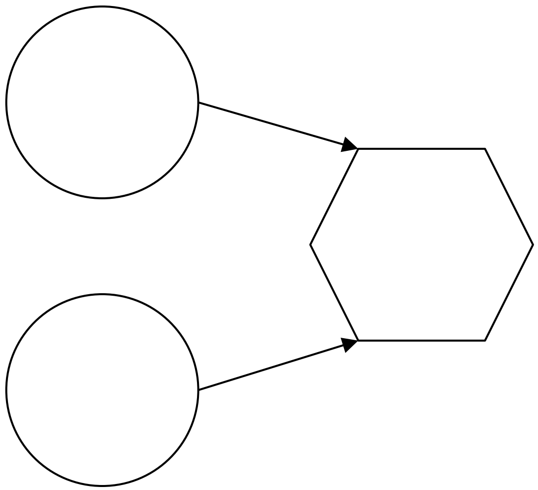
Components of a Decision

1. Decisions □
   - A choice between two or more alternatives that involves an irrevocable allocation of resources
   - e.g. invest/do not invest in new advanced weapon system
2. Uncertainties ○
   - Events that can occur, probabilistically with different degrees, which affect the outcome
   - e.g. mission success, readiness of technology, adversary countermeasures
3. Deterministic Nodes ◎
   - If all inputs to a deterministic node are known, then there is no longer any uncertainty about it
   - e.g. net income if revenue and expenses are known
4. Outcomes (symbol varies, sometimes ◁ or ⬡)
   - Quantitative representation of the value of each combination of decision and uncertainty
   - e.g. monetary value, expected casualties

Influence Diagrams

- Conditional dependence
  - Arrow between two uncertainties
  - Arrow from decision node to uncertainty
  - Absence of arrow between two nodes asserts independence
- Inputs to deterministic node
  - Arguments of a function
- Prior knowledge of an uncertainty or event
  - Arrow from any node into a decision indicates that the result of the node are known before decision is made
  - If the parent node of a decision is an uncertainty, that uncertainty has been resolved by the time of the decision
  - If the parent node is another decision, the order of the decisions is made explicit



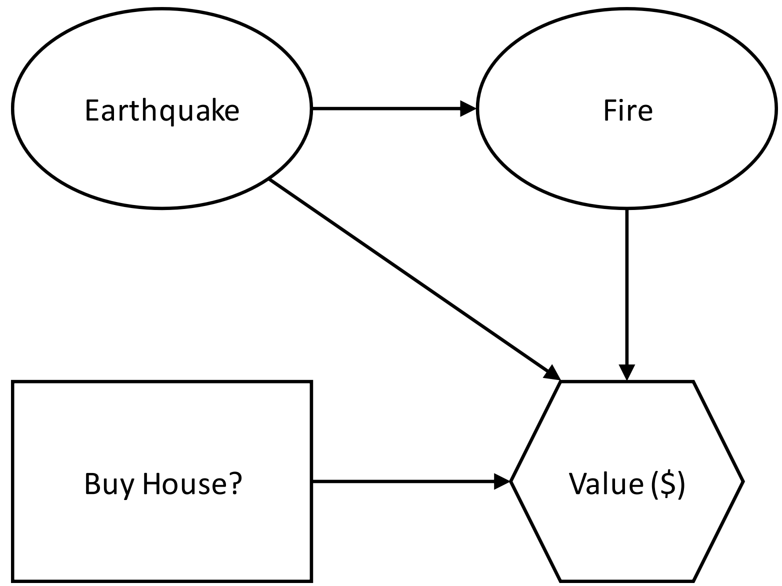


Influence Diagram Example

- Scenario: Suppose there is an historic house for sale in San Francisco with an amazing view of the Bay. The asking price for this house is $3 million. Conditional on the house not getting destroyed by an Act of God, John would be willing to pay up to $4 million for the house. John believes that there are two kinds of Acts of God which might destroy the house: earthquakes and fires. If an earthquake occurs, even if it doesn’t destroy the house directly, it increases the probability of a fire, which may damage or destroy the house.
- Question: What is the influence diagram which reflects this scenario?

Influence Diagram Example

- Answer:

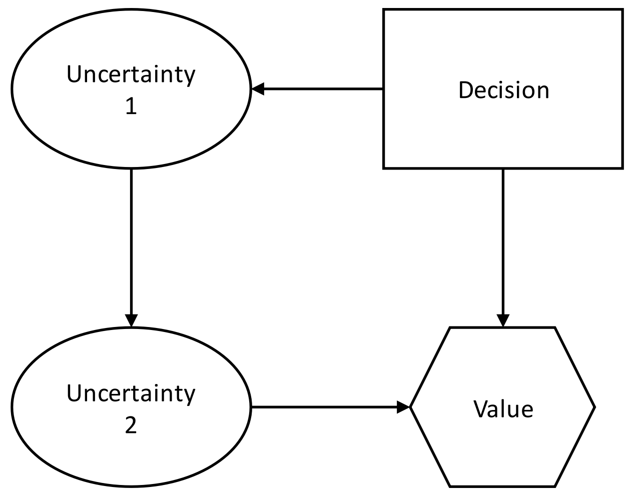


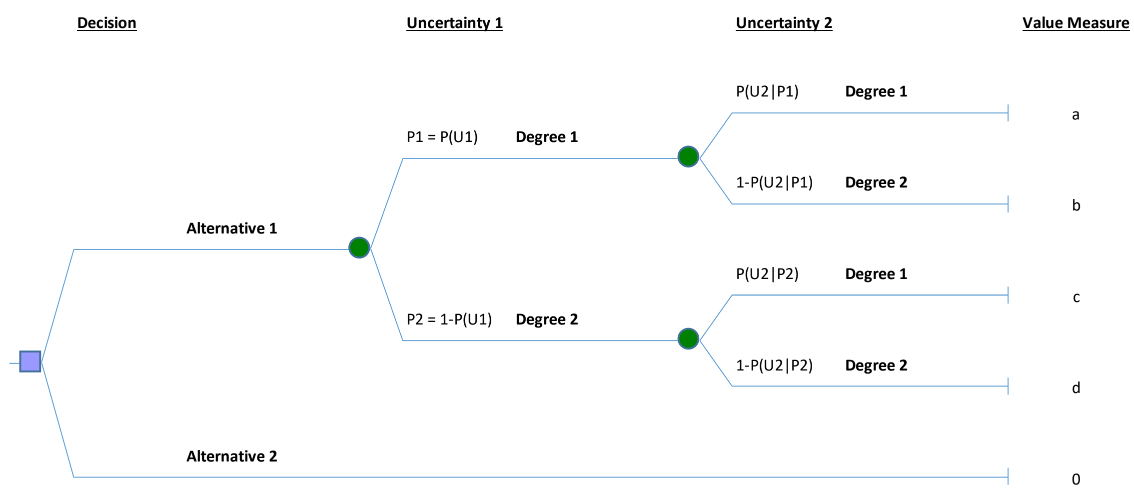
- Earthquake⟶Fire:
  - Probability of fire depends on whether or not an earthquake has occurred
- BuyHouse⟶$, Earthquake⟶$, Fire⟶$:
  - Value that we “book” by making the decision depends on (a) the result of our decision, and (b) whether or not the house is subsequently destroyed
- Why no arrows from BuyHouse to Earthquake or Fire?

Decision Trees and Probability Trees

- Scenarios depicted in influence diagrams can also be depicted in decision trees
- If no information about any uncertainties is known at the time of a decision, the decision is usually drawn as the root node of the decision tree
  - The branches of the node are the different alternatives that could be selected
  - Each subsequent node is an uncertainty (along with its possible realizations) or a subsequent decision
- If an uncertainty precedes a decision in an event tree, then the uncertainty is resolved and is known to the decision maker
  - Only time that **time** explicitly plays a role in decision trees
- The leaves of the tree represent all possible realizations of the joint distribution over the uncertainties together with the decision(s)
  - Leaves represent the value that the decision maker places on that specific outcome
- **Probability trees are just decision trees, but without a decision node**
  - End leaves depict joint probability distribution as the values

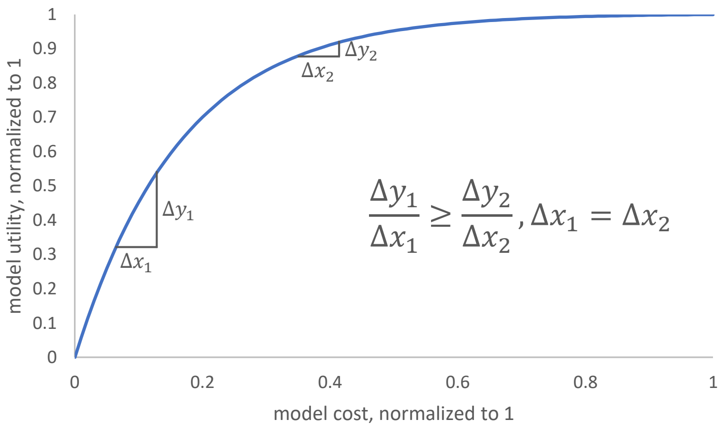
Influence Diagrams and Decision Trees





Model Building: Cogency Versus Verisimilitude

- Model building adheres to economic law of diminishing marginal utility
- Key distinction in models: cogency versus verisimilitude
  - Entails an element of art in model building



We should strive to build cogent models, not high verisimilitude models

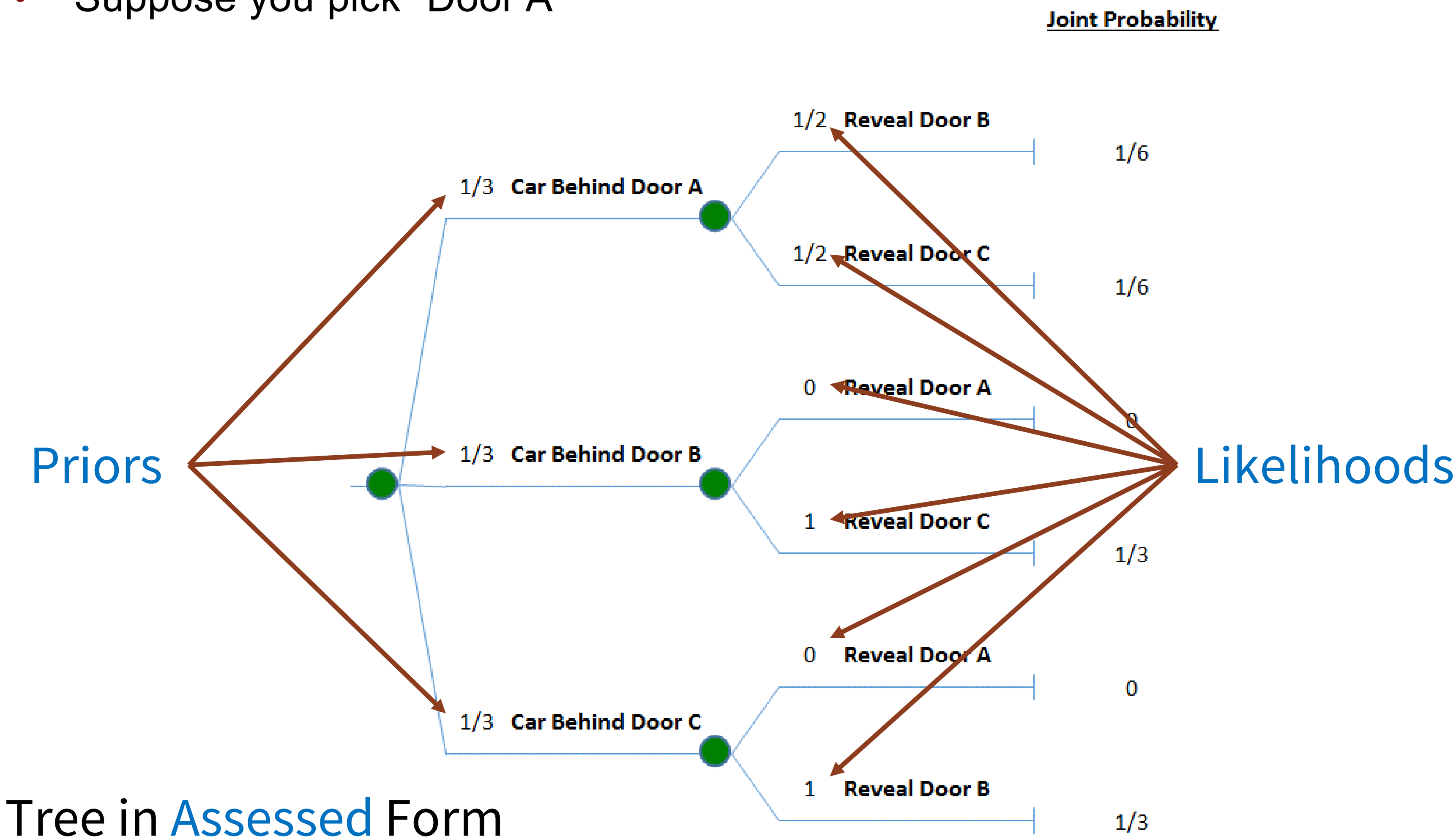
Cogency Versus Verisimilitude

- “Finally, we want to say a few words about model complexity. In our experience, companies tend to build too much detail into their decision-making models from the start and focus too much energy on specific cases or particular inputs such as operating costs. The distinction we would like to suggest is cogency versus verisimilitude. Cogency is having the property of being compelling. Verisimilitude means being true to life. We seek cogent decision models, rather than models exhibiting verisimilitude. Perhaps an analogy will help. Consider model-train building. What makes a very good model train? One that is true to life. For example, an average model train might include a bar car. A good model train might include a bar car with people. A very good model train might show the people holding drinks. In an award-winning model train, you would be able to tell what kind of drinks the people were holding – for example, martinis with olives. Yet, inside a world-class model train, you would be able to see the pimento inside the olive! Decision analysis is not about building model trains. Rather we seek to provide insights into decision makers regarding critical decisions. Rarely does this require “pimento-level” modeling detail…Building in detail is easy. Building in incisiveness is hard work. As Howard (1980) has written, ‘…the real problem in decision analysis is not making analyses complicated enough to be comprehensive, but rather keeping them simple enough to be affordable and useful.’”

E. Bickel and R. Bratvold, “Decision-Making in the Oil and Gas Industry--From Blissful Ignorance to Uncertainty-Induced Confusion,” SPE International, 2007.

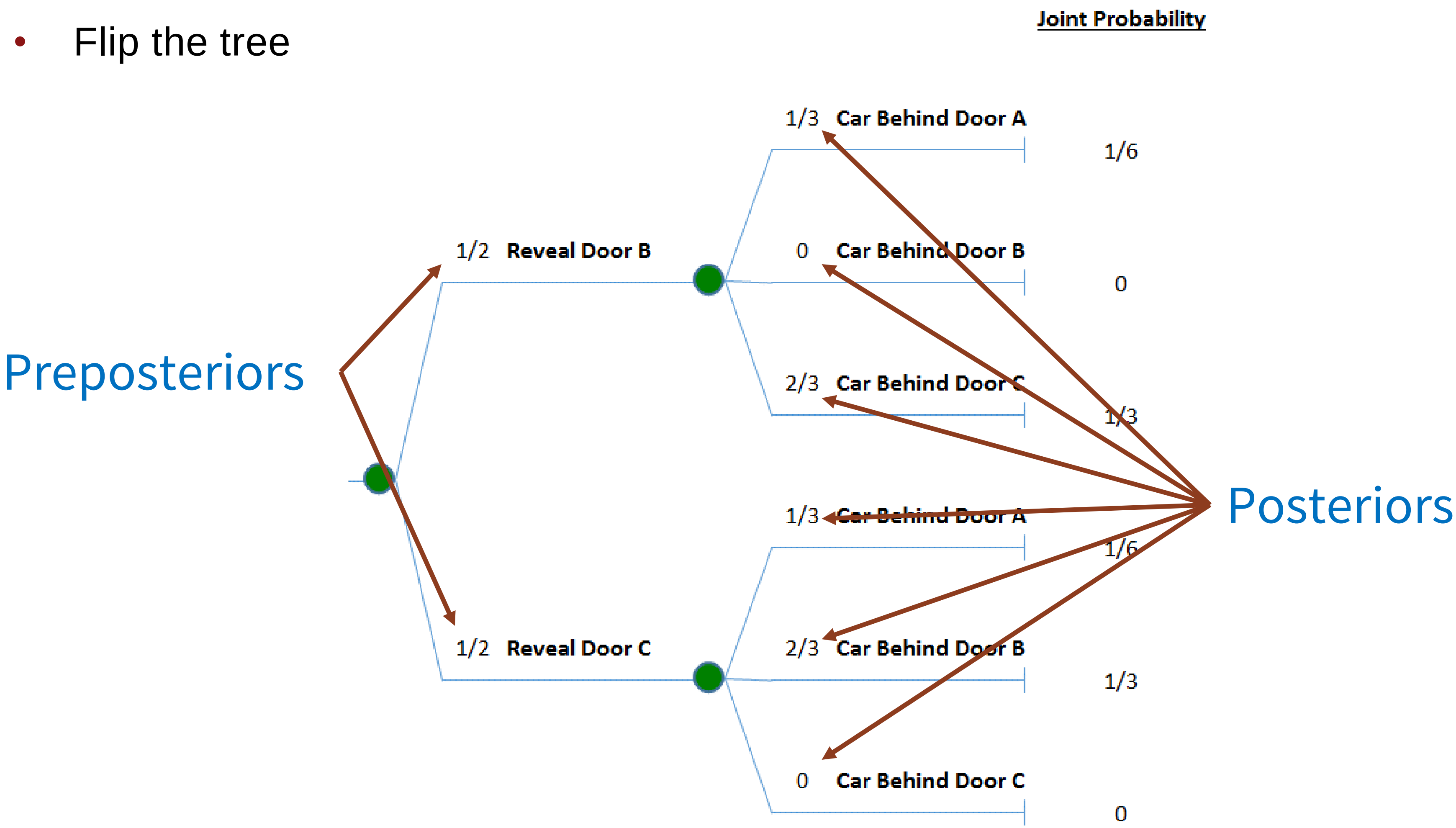
Monty Hall Problem Solution Using Trees

- Suppose you pick “Door A”



Monty Hall Problem Solution Using Trees

- Flip the tree



Tree in Inferential Form

Bayes' Rule tells us we should *always* switch doors!